香港城市大學
City University of Hong Kong

fSuSiE enables fine-mapping of QTLs from genome-scale molecular profiles

Speaker: Yuekai Li

Major: Biostatistics

CONTENTS

1. Background

2. Methods

3. Simulations

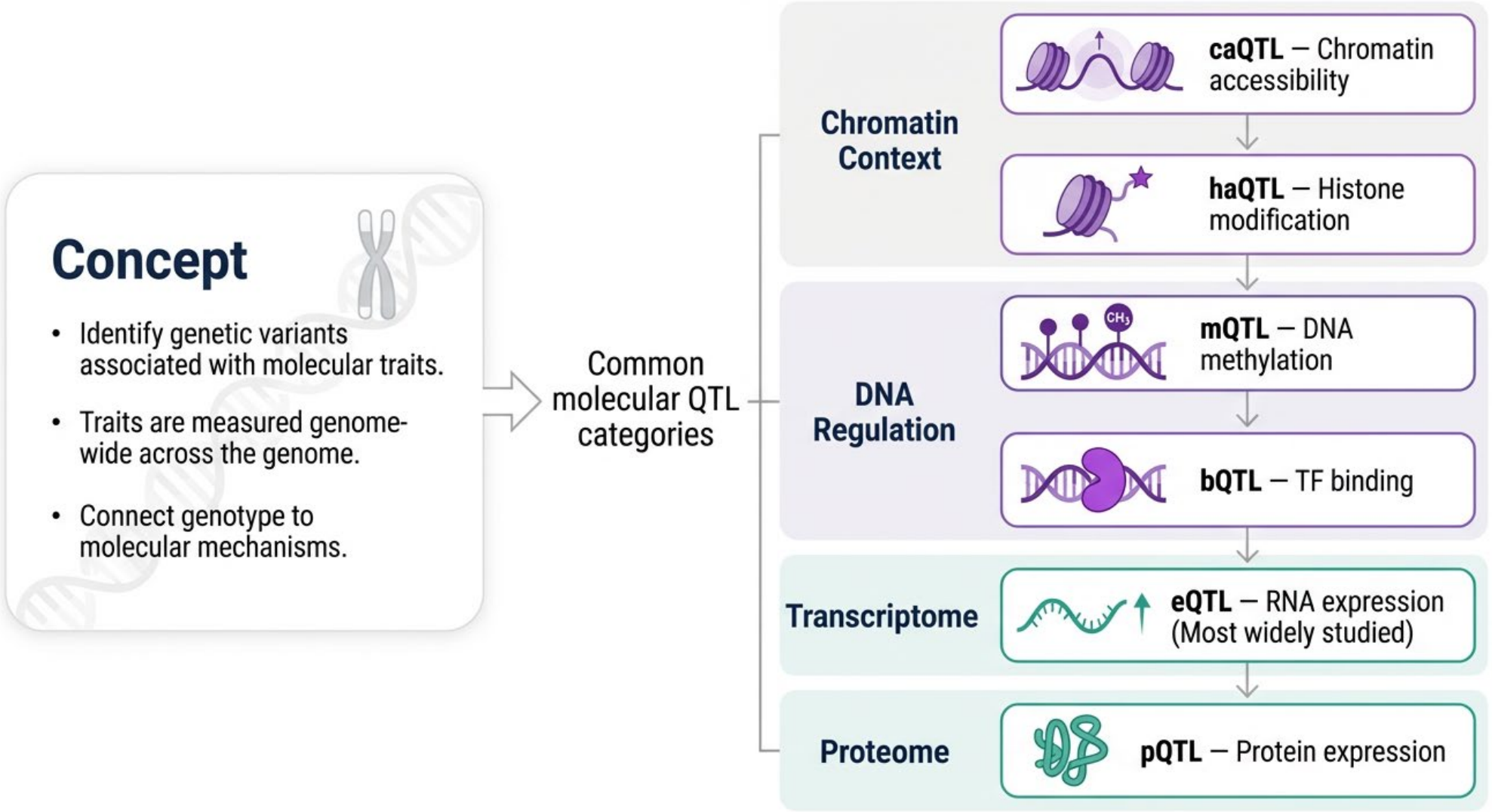
4. Real-data applications

香港城市大學
City University of Hong Kong

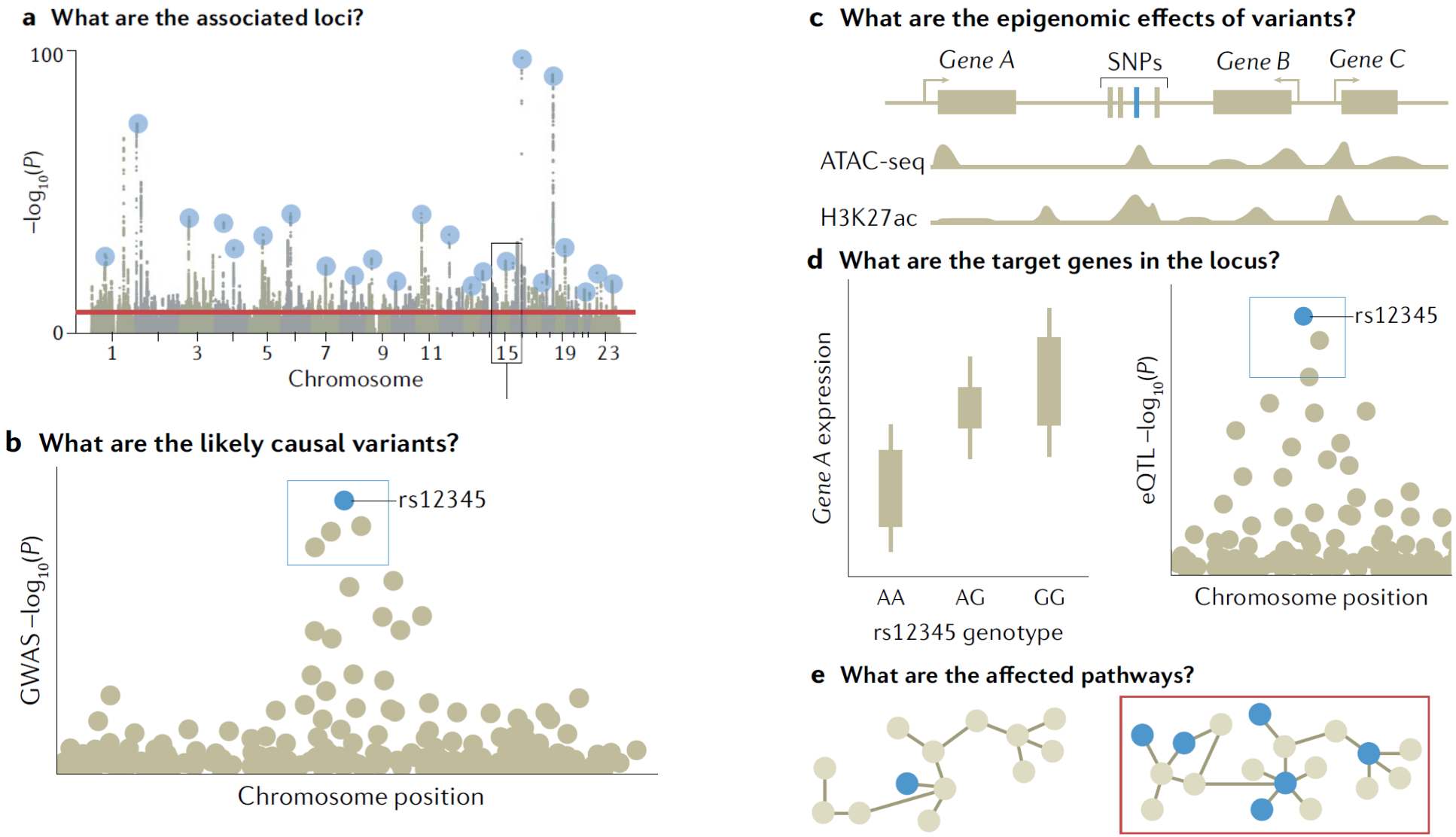
01

Background

Molecular quantitative trait locus (QTL) mapping



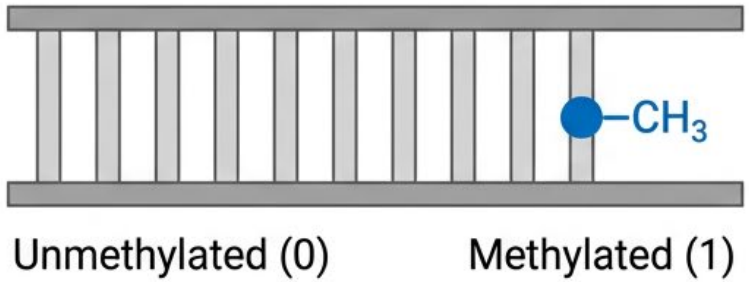
Fine-mapping and downstream analysis



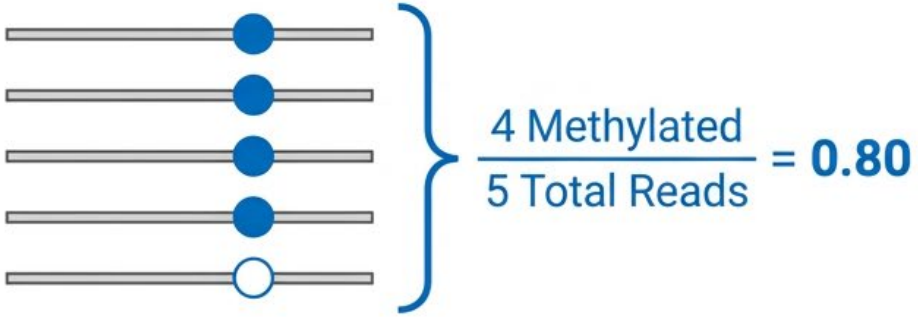
DNA methylation and mQTL analysis

DNA Methylation

Methylation Status



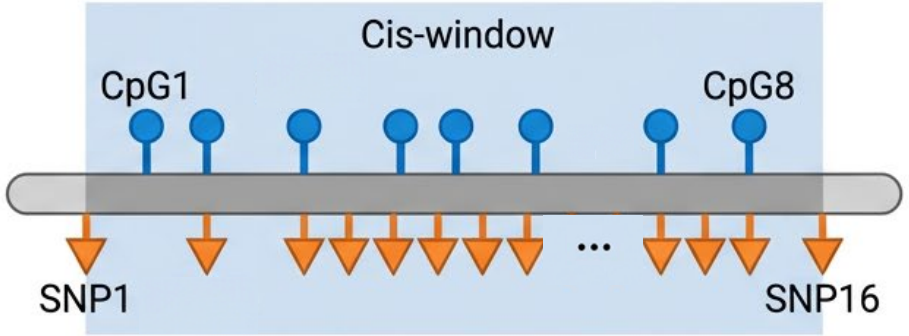
From Reads to Levels



mQTL analysis

Methylation Y

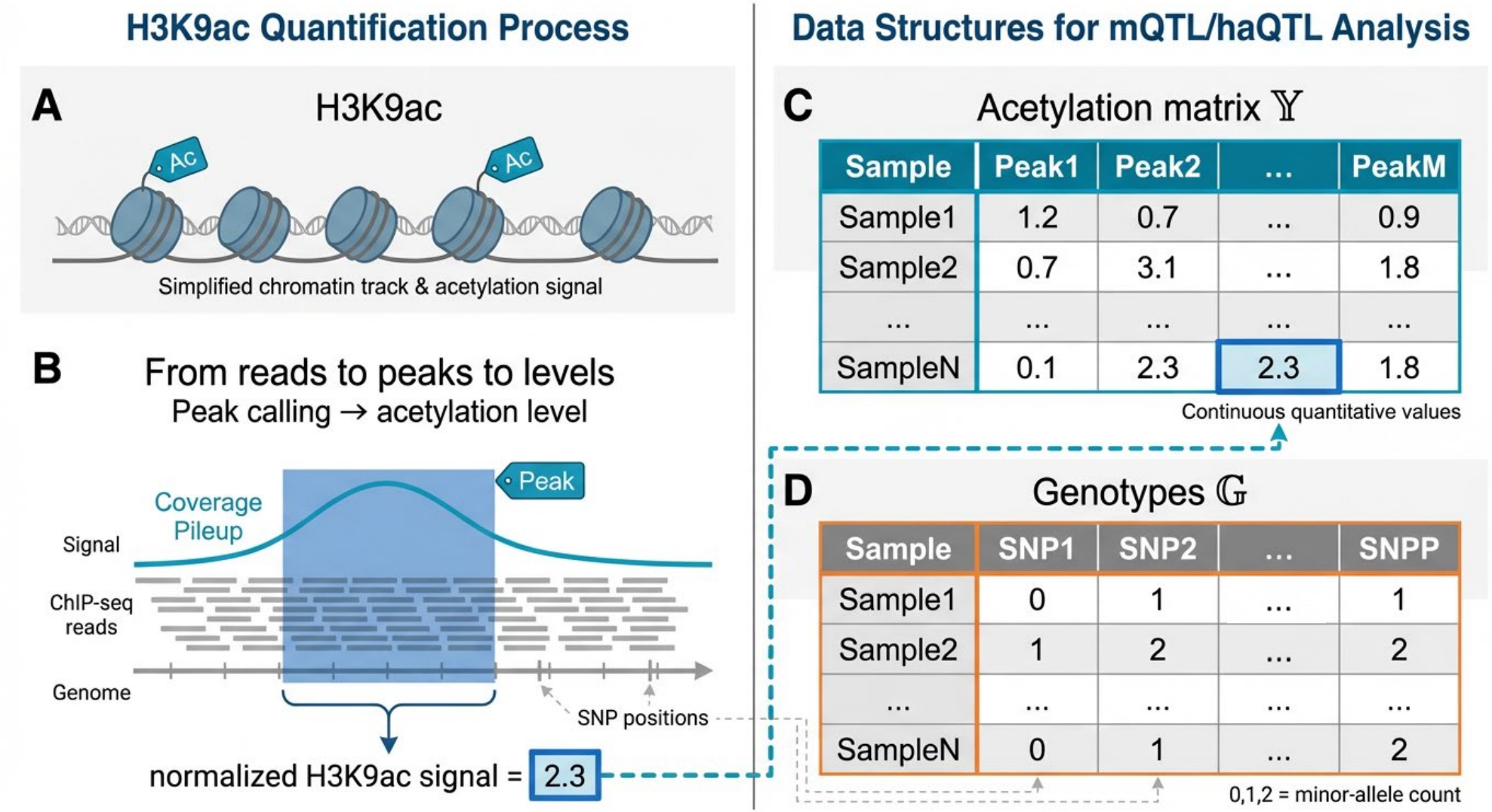
	CpG1	CpG2	...	CpG8
Sample1	0.05	0.80	...	0.67
Sample2	0.10	0.99	...	0.52
...	...	...	...	...
SampleN	0.05	0.88	...	0.67



Genotypes G

	SNP1	SNP2	...	SNP16
Sample1	0	2	1	0
Sample2	0	2	1	0
...	...	...	...	...
SampleN	0	2	1	0

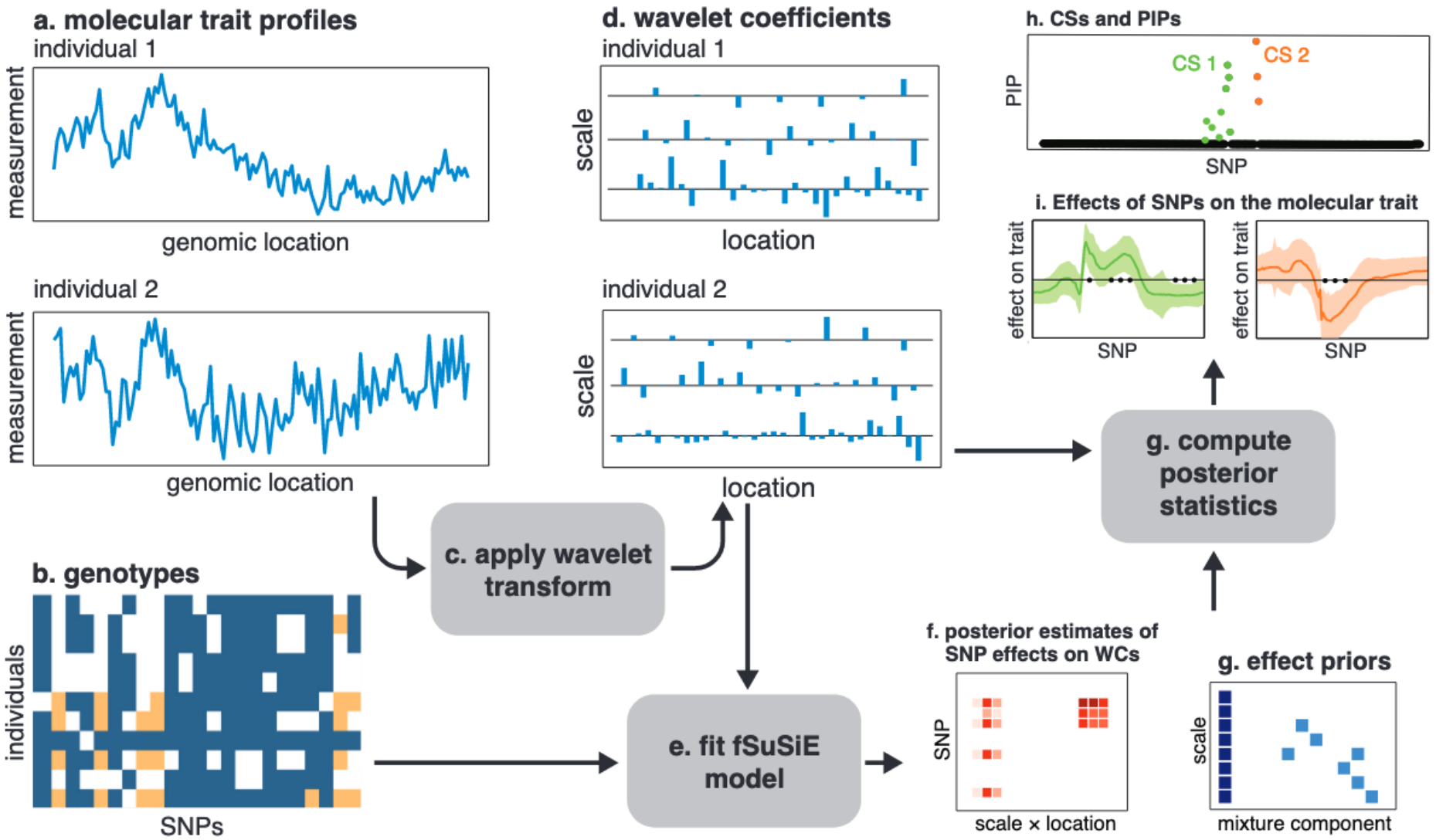
Histone acetylation and haQTL analysis



Sample	Peak1	Peak2	...	PeakM
Sample1	1.2	0.7	...	0.9
Sample2	0.7	3.1	...	1.8
...	...	...	...	...
SampleN	0.1	2.3	2.3	1.8

Sample	SNP1	SNP2	...	SNPP
Sample1	0	1	...	1
Sample2	1	2	...	2
...	...	...	...	...
SampleN	0	1	...	2

Overview of fine-mapping of molecular traits using fSuSiE



香港城市大學
City University of Hong Kong

02

Methods

Haar wavelet transform

For equally-spaced spatial observations: $\boldsymbol{f} = (f_1, f_2, \dots, f_T)$, with $T = 2^S$,

Wavelet coefficient (WC) $d_{s,l}$:

The index s is the "scale" of a wavelet coefficient and index l is the wavelet location.

WCs decompose the variation of $\boldsymbol{f}$ across multiple resolution levels ($s = 0, 1, 2, \dots, S$):

$$\begin{cases} d_{0,1} = \sum_{t=1}^{T} f_t, \\ d_{1,1} = \sum_{t=1}^{T/2} f_t - \sum_{t=T/2+1}^{T} f_t, \\ d_{2,1} = \sum_{t=1}^{T/4} f_t - \sum_{t=T/4+1}^{T/2} f_t, \\ d_{2,2} = \sum_{t=T/2+1}^{3T/4} f_t - \sum_{t=3T/4+1}^{T} f_t, \\ \vdots = \end{cases}$$

Scale $s = 0$: 1 coefficient,

Scale $s = 1,2, \dots, S-1$,: $2^s - 1$ coefficients.

Total: $1 + \sum_{s=1}^{S} 2^{s-1} = 2^S = T$

The total number of WCs exactly equals the data length.

An example of Haar wavelet transform

$$\begin{array}{cccccccc} f_1, & f_2, & f_3, & f_4, & f_5, & f_6, & f_7, & f_8 \end{array}$$

$$\boldsymbol{f} = (9, 8, 7, 6, 9, 8, 3, 2)$$

Haar wavelet transform

$$\boldsymbol{d} = (52, 8, 4, 12, 1, 1, 1, 1)$$

$$d_{0,1}, d_{1,1}, d_{2,1}, d_{2,2}, d_{3,1}, d_{3,2}, d_{3,3}, d_{3,4}$$

Equivalent form of matrix multiplication:

$$\begin{pmatrix} 9 \\ 8 \\ 7 \\ 6 \\ 9 \\ 8 \\ 3 \\ 2 \end{pmatrix}^T \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 52 \\ 8 \\ 4 \\ 12 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}^T$$

measurements $\boldsymbol{f}^T$ projection matrix $\mathbf{W}$ WCs $\boldsymbol{d}^T$

$$\boxed{\boldsymbol{f}^T \mathbf{W} = \boldsymbol{d}^T}$$

■ Linear interpolation procedure for non-decimated and unevenly-spaced data

The approach of Kovac and Silverman for non-decimated (size $\neq 2^S$) and unevenly-spaced data consist on interpolating the observed data on a evenly spaced grid of length 2^S.

Suppose, we observe $f(\tau_i)$ at T' unevenly spaced time points $\tau_0, \dots, \tau_{T'}$, and consider a new set of evenly space point $t_0, \dots, t_T$, where $T = 2^S$ and $S = \min\{s \in N, 2^s \geq T'\}$:

$$f_{inter}(t_j) = \begin{cases} f(\tau_0) & \text{if } t_j \leq \tau_0 \\ f(\tau_i) + (t_j - \tau_i)\dfrac{f(\tau_{i+1}) - f(\tau_i)}{\tau_{i+1} - \tau_i} & \text{if } \tau_i \leq t_j \leq \tau_{i+1} \\ f(\tau_{T'}) & \text{if } t_j \geq \tau_{T'} \end{cases}$$

The expression can then be written in terms of the matrix as follows:

$$\boldsymbol{f}_{inter}^T = \boldsymbol{f}^T \boldsymbol{\Omega}$$

The model of Sum of Single Effects (SuSiE)

We relate the phenotype (a single univariate trait) $\mathbf{y}$ to genotypes $\mathbf{X}$ with the linear model, and we consider the following sum-of-single-effects structure:

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}, \mathbf{e} \sim \mathcal{N}_n\left(\mathbf{0}, \sigma^2 \mathbf{I}_n\right)$$
$$\mathbf{b} = \sum_{l=1}^{L} \boldsymbol{\gamma}_l \cdot b_l, b_l \sim \mathcal{N}_1\left(0, \sigma_{bl}^2\right), \boldsymbol{\gamma}_l \sim \mathrm{Mult}(1, \boldsymbol{\pi})$$

- $\mathbf{b} \in \mathbf{R}^p$ is the sparse vector of SNP effect sizes; b_l is the effect size of the l-th causal signal;
- $\boldsymbol{\gamma}_l$ is a binary vector with $\gamma_{lj} = 1$ indicating the l-th causal signal is attributed to the j-th SNP;
- $\boldsymbol{\pi}$ is the vector of prior causal probabilities with $\pi_j = 1/p$.

Advantages:

The variational "sum of single effects" assumption treats causal signals as independent, giving simple updates and credible sets.

By iteratively adding single-effect components, the model can capture multiple causal variants even in complex LD patterns.

Variational EM makes it faster than MCMC, and the framework is straightforward to extend with additional priors or phenotypes.

The model of functional SuSiE (fSuSiE)

Next, we consider a multiple linear regression model: $\mathbf{Y} = \mathbf{XB} + \mathbf{E}^{\mathrm{m}}$

where $\mathbf{Y}$ is an $N \times T$ matrix of phenotypes, $\mathbf{X}$ is an $N \times J$ matrix of genotypes, and $\mathbf{B}$ is the $J \times T$ matrix of effects b_{jt} to be estimated. The corresponding multiple wavelet regression model is:

$$\boxed{\begin{aligned} &\mathbf{D} = \mathbf{XF} + \mathbf{E}, \mathbf{E}_{it} \sim \mathcal{N}\left(0, \sigma^2\right) \\ &\mathbf{F} = \sum_{r=1}^{R} \mathrm{diag}\left(\boldsymbol{\gamma}^{(r)}\right) \times \mathbf{F}^r, \boldsymbol{\gamma}^{(r)} \sim \mathrm{Multinom}(1, \boldsymbol{p}), f_{jt}^r \sim g_t^r \end{aligned}}$$

- $\mathbf{D} = \mathbf{YW}$ is an $N \times T$ matrix of transformed data (called "wavelet coefficients");
- $\mathbf{F} = \mathbf{BW}$ is the sparse matrix of effects f_{jt} in the wavelet space;
- $\gamma^{(r)}$ is a binary vector with $\gamma_{rj} = 1$ indicating the r-th causal signal is attributed to the j-th SNP;
- $\boldsymbol{p}$ is the vector of prior causal probabilities with $\boldsymbol{p}_j = 1/J$.

The scale-dependent mixture priors for fSuSiE

$$\mathbf{D} = \mathbf{X}\mathbf{F} + \mathbf{E}, \mathbf{E}_{it} \sim \mathcal{N}(0, \sigma^2)$$
$$\mathbf{F} = \sum_{r=1}^{R} \mathrm{diag}\left(\boldsymbol{\gamma}^{(r)}\right) \times \mathbf{F}^r, \boldsymbol{\gamma}^{(r)} \sim \mathrm{Multinom}(1, \boldsymbol{p}), f_{jt}^r \sim g_t^r$$

The shrinkage-per-scale (SPS) prior:

$$g_t^r = \sum_{k=0}^{K} \pi_{k,s}^r \mathcal{N}(0, \sigma_{k,s}^2)$$

a special case →

the mixture proportions and variances do not depend on the scale s

The independent shrinkage (IS) prior:

$$g_t^r = \sum_{k=0}^{K} \pi_k^r \mathcal{N}(0, \sigma_k^2)$$

- $\pi_{ks}^r \geq 0$ denote the mixture weights for each single function $r = 1, \ldots, R$, with $\sum_{k=0}^{K} \pi_{ks}^r = 1$.

  π_{ks}^r are estimated and depend on the scale s.

- $\sigma_{ks}^2 \geq 0$ denote the variances of the mixture components, with $0 \leq \sigma_{1,s}^2 < \cdots < \sigma_{K-1,s}^2 < \sigma_{Ks}^2$

  The number of mixture components K and the component variances σ_{ks}^2 are fixed, user-specified quantities.

Algorithms for fitting the fSuSiE model

Algorithm 1: Basic Iterative Bayesian stepwise selection for functional data

Input
Data : $\boldsymbol{D}, \boldsymbol{X}$
Prior :G
A function: $SFR(\boldsymbol{X}, \boldsymbol{D}, G, \sigma_0^2) \rightarrow (\alpha, \tilde{\boldsymbol{F}}^*)$
Initialization
$\boldsymbol{R}_r \longleftarrow \boldsymbol{D}$
repeat
for $r = 1$ **to** R **do**
  $\boldsymbol{R}_r \leftarrow \mathbf{D} - \mathbf{X} \sum_{r' \neq e} \hat{\boldsymbol{F}}^{(r')}$
  $(\boldsymbol{\alpha}_e, \tilde{\boldsymbol{F}}^r, (\sigma_{e,j,s,l})) \longleftarrow SFR(\boldsymbol{X}, \boldsymbol{R}_r, G, \sigma_0^2)$
  $\hat{\boldsymbol{F}}^{(r)} \longleftarrow \tilde{\boldsymbol{F}}^r \odot \boldsymbol{\alpha}_e$
until the convergence criterion is satisfied
Output

Algorithms for fitting the fSuSiE model

Factorization assumption:

$$q\left(\mathbf{F}^{(1)},\ldots,\mathbf{F}^{(R)}\right)=\prod_{r=1}^{R} q_r\left(\mathbf{F}^{(r)}\right)$$

Algorithm 1: Basic Iterative Bayesian stepwise selection for functional data

Input
Data : $\boldsymbol{D}, \boldsymbol{X}$
Prior :G
A function: $SFR(\boldsymbol{X}, \boldsymbol{D}, G, \sigma_0^2) \rightarrow (\alpha, \tilde{\boldsymbol{F}}^*)$
Initialization
$\boldsymbol{R}_r \longleftarrow \boldsymbol{D}$
repeat
for $r = 1$ **to** R **do**
  $\boldsymbol{R}_r \leftarrow \mathbf{D} - \mathbf{X}\sum_{r' \neq e} \hat{\boldsymbol{F}}^{(r')}$
  $(\boldsymbol{\alpha}_e, \tilde{\boldsymbol{F}}^r, (\sigma_{e,j,s,l})) \longleftarrow SFR(\boldsymbol{X}, \boldsymbol{R}_r, G, \sigma_0^2)$
  $\hat{\boldsymbol{F}}^{(r)} \longleftarrow \tilde{\boldsymbol{F}}^r \odot \boldsymbol{\alpha}_e$
until the convergence criterion is satisfied
Output

IBSS is a coordinate ascent algorithm for maximizing the ELBO F over q satisfying the approximation.

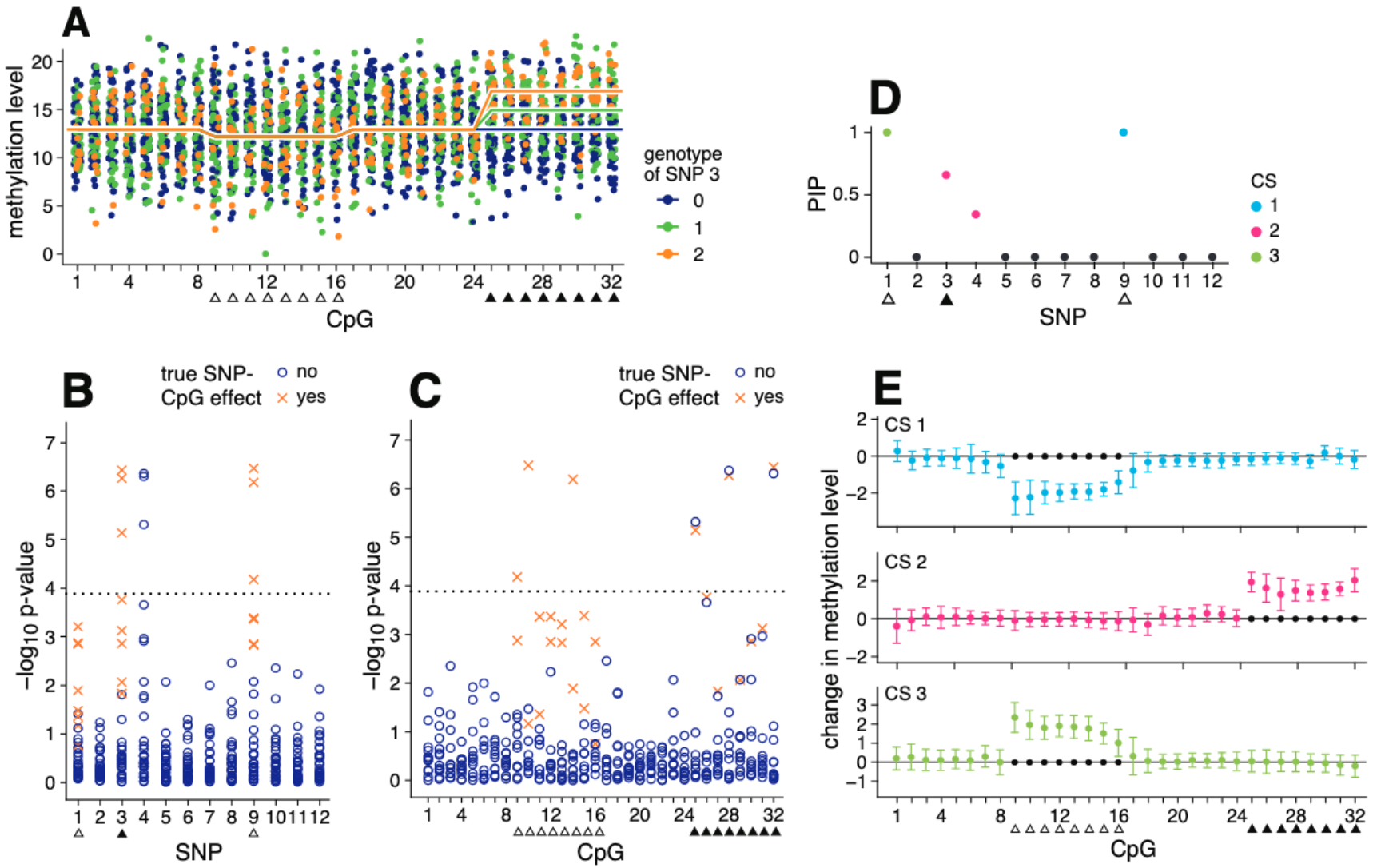
Credible sets, sentinel SNPs and credible bands

Credible set (CS): a level-ρ CS is defined as a set of SNPs that is as small as possible such that it has probability at least ρ of containing an effect SNP. The number of CSs should reflect the number of causal SNPs.

Sentinel SNP: Within each CS, we refer to the SNP with the highest PIP as the "sentinel SNP".

Credible band: For each sentinel SNP, a pointwise credible band that is designed to have high probability of containing the effect at each location. If the credible band for a particular location excludes zero then we say that the CS containing that sentinel SNP is "significant" for this location.

A simulated example for Molecular QTL mapping with fSuSiE




```
F <- matrix(0,p,m)
F[1,9:16] <- 2.3
F[9,9:16] <- (-2.3)
F[3,25:32] <- 2
```

香港城市大學
City University of Hong Kong

03

Simulations

Data simulation for SNP genotypes

- The fine-mapping regions for the simulations were selected uniformly at random from **94 breast cancer loci** (median: ~1 Mb) on autosomal chromosomes.

- We simulated genotypes with **realistic patterns of linkage disequilibrium** (LD) using **sim1000G***, with genotype data from the 1000 Genomes Phase 3 whole-genome sequencing as the input source.

- For a single simulation, the genotype matrix, **X**, was a matrix with **N = 100 rows** (individuals) and J columns (SNPs), in which J ranged from approximation 1,500 to 4,000.

*: Dimitromanolakis, A., J. Xu, A. Krol, and L. Briollais (2019). sim1000G: a user-friendly genetic variant simulator in R for unrelated individuals and family-based designs. BMC Bioinformatics 20, 26.

■ Whole-genome bisulfite sequencing (WGBS) simulation for CpG methylation levels

The methylation profiles, **Y**, were simulated using the following logistic regression model:

$$y_{it} = \text{sigmoid}(z_{it}),$$
$$z_{it} = \sum_{j=1}^{J} x_{ij}\, b_{jt} + \text{probit}(\beta_t) + e_{it},$$
$$e_{it} \sim N(0, \sigma^2).$$

- β_t: a baseline methylation level (a "Beta value") simulated using OmicsSIMLA.
- probit(x): probit link function, which is the inverse function of cumulative density function of N(0, 1).
- $\boldsymbol{b}_j \in N_T(0, \mathbf{C})$: The effects of the causal SNP j, in which **C** is a T × T correlation matrix.

We considered two different designs of **C** that resulted in two different sets of WGBS simulations, which we referred to as "WGBS block" and "WGBS decay" in the text.

Three epigenetic mechanisms in simulation

01

WGBS block

"Constant changes in methylation across contiguous windows, with abrupt changes between windows."

$$\mathbf{C}_{\text{block}} = \begin{bmatrix} \mathbf{J}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{J}_3 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{J}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{J}_3 \end{bmatrix}, \quad \mathbf{J}_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

02

WGBS decay

"Changes that varied smoothly within the locus."

$$\mathbf{C}_{\text{decay}} = \begin{bmatrix} \mathbf{R}_3(\rho) & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_3(\rho) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{R}_3(\rho) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{R}_3(\rho) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{R}_3(\rho) \end{bmatrix}$$

$$\mathbf{R}_3(\rho) = \begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix}$$

03

Wavelet simulations

"To benchmark fSuSiE in the ideal setting, we performed simulations in which the methylation levels were simulated from the wavelet model assumed by fSuSiE."

Methods compared in the simulations

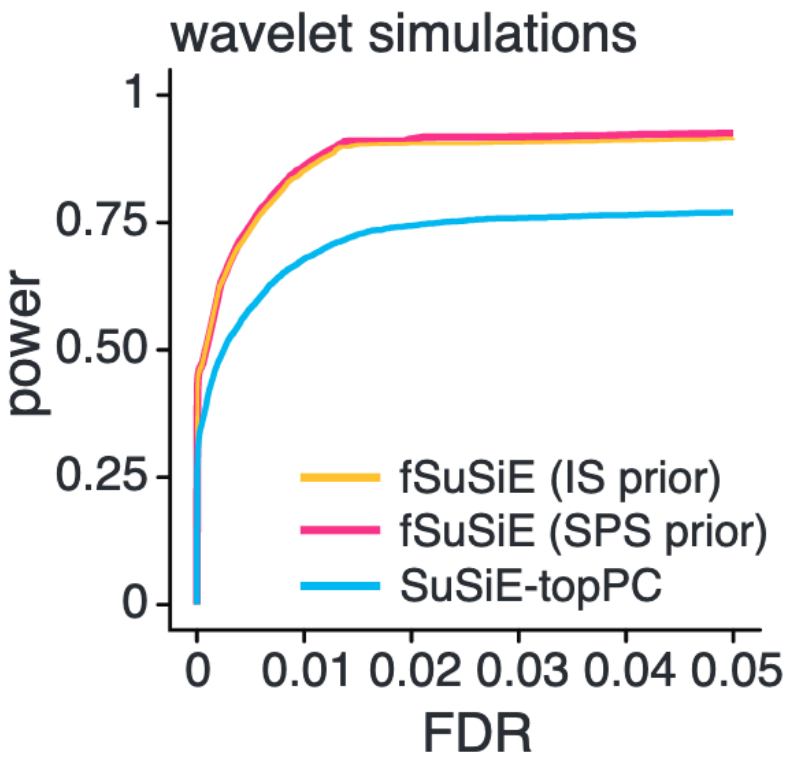
fSuSiE: the first available tool to perform statistical fine-mapping of functional traits.

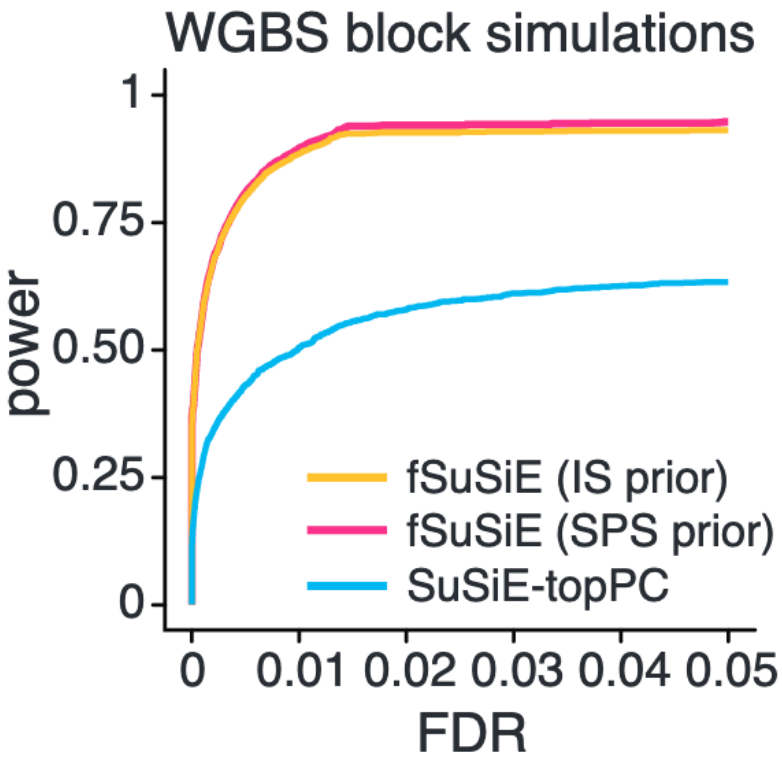
SuSiE-topPC: running SuSiE on the top PC computed from the column-centered **Y**.

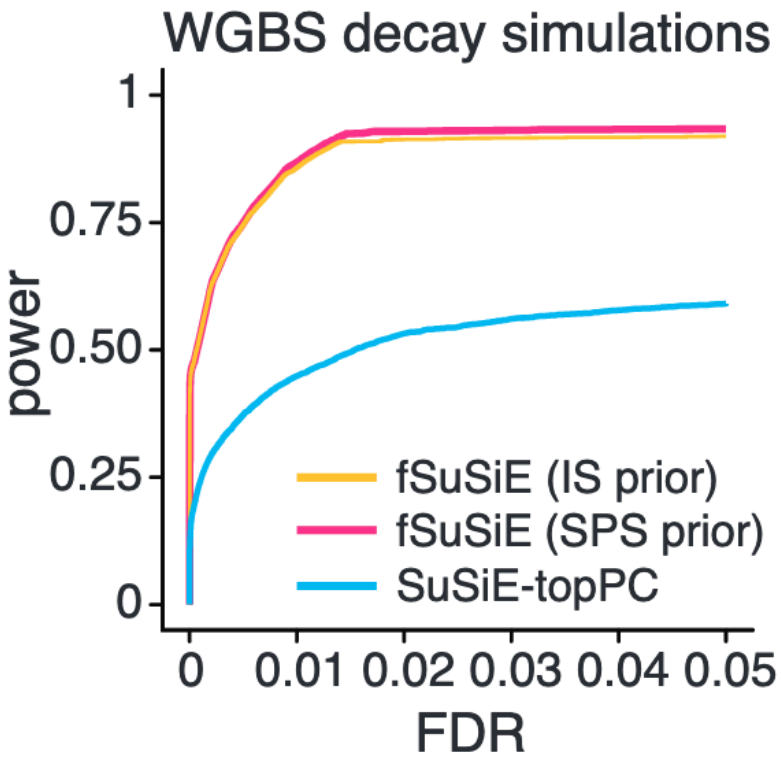
SuSiE-topPC has important limitations compared with **fSuSiE**:

- In real applications the top PC may primarily capture non-genetic effects such as experimental batch effects or population structure.
- It only identifies causal SNPs that impact some aspect of the high-dimensional trait, without identifying which specific molecular locations (CpGs) are affected by these SNPs.

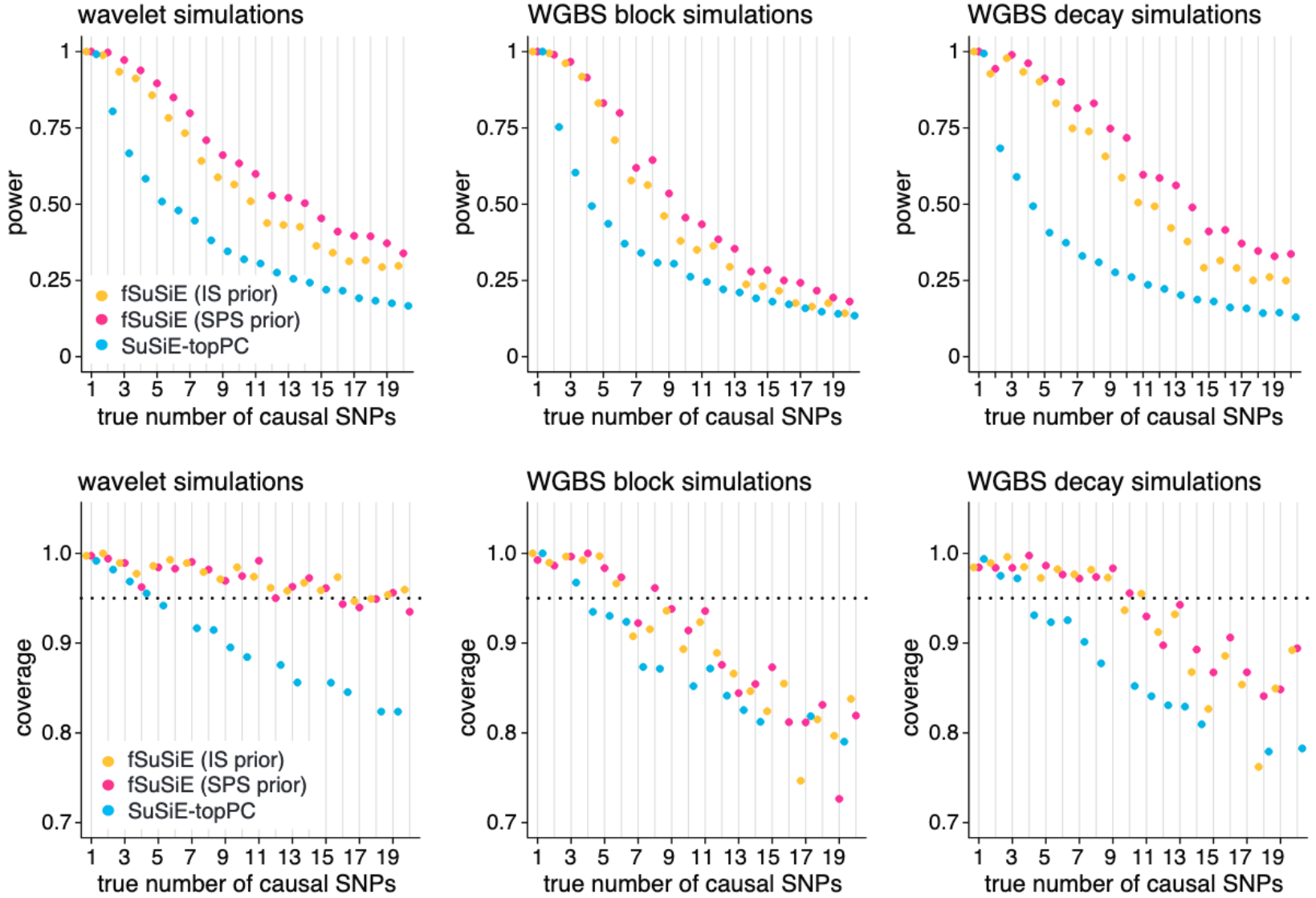
Performance assessment of detection of SNPs (PIPs)



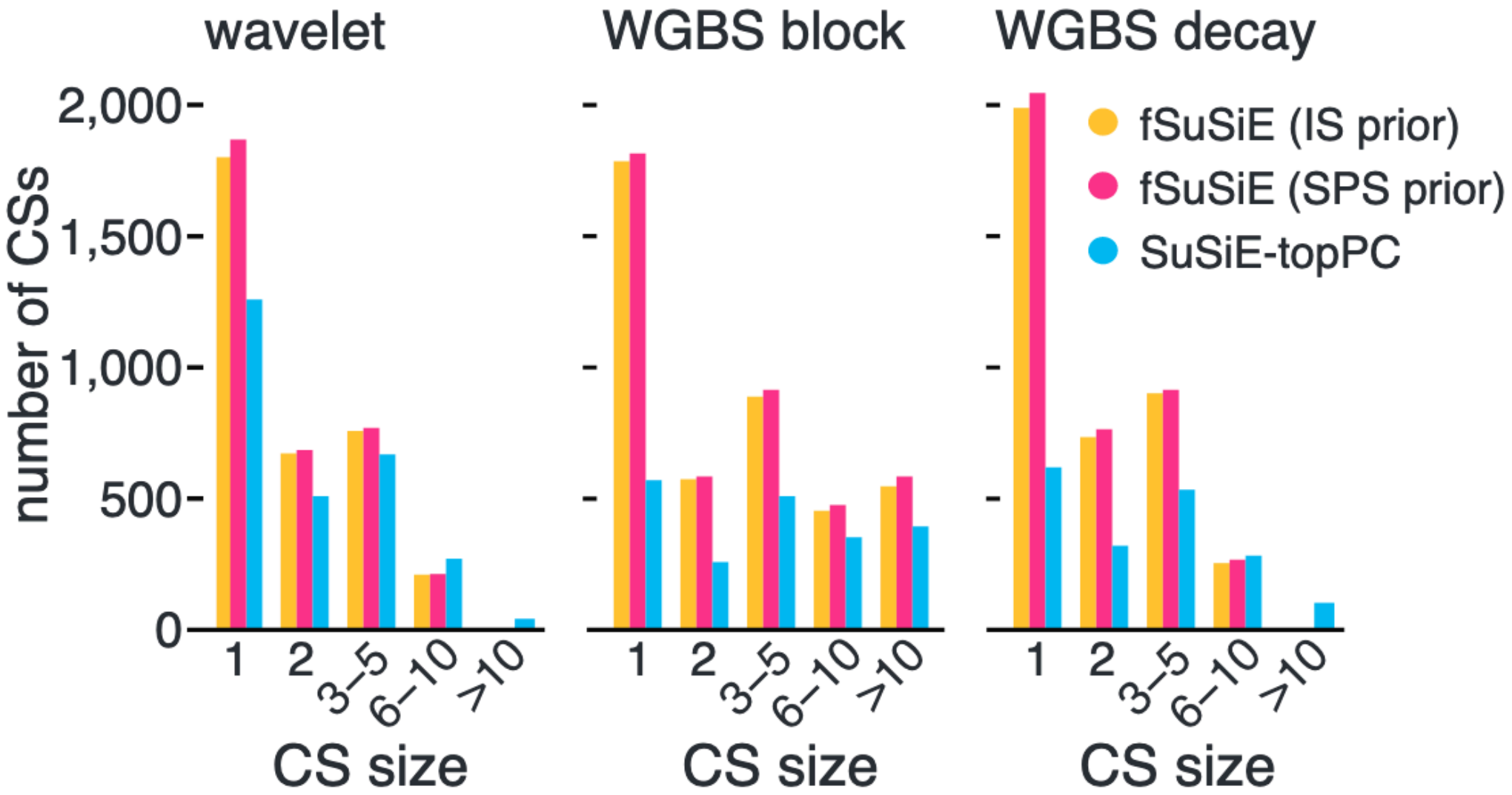




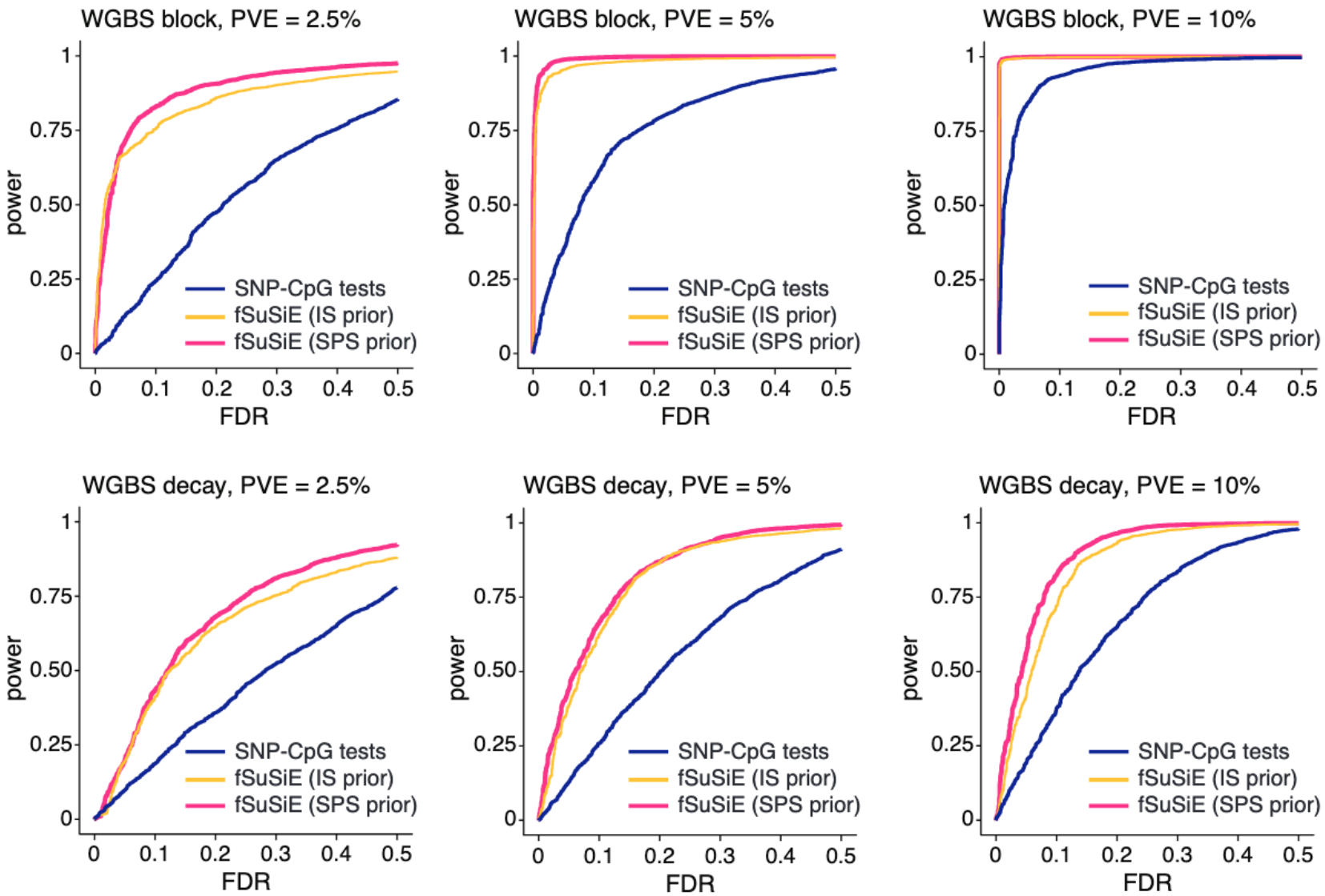
Performance assessment of detection of SNPs (CSs)



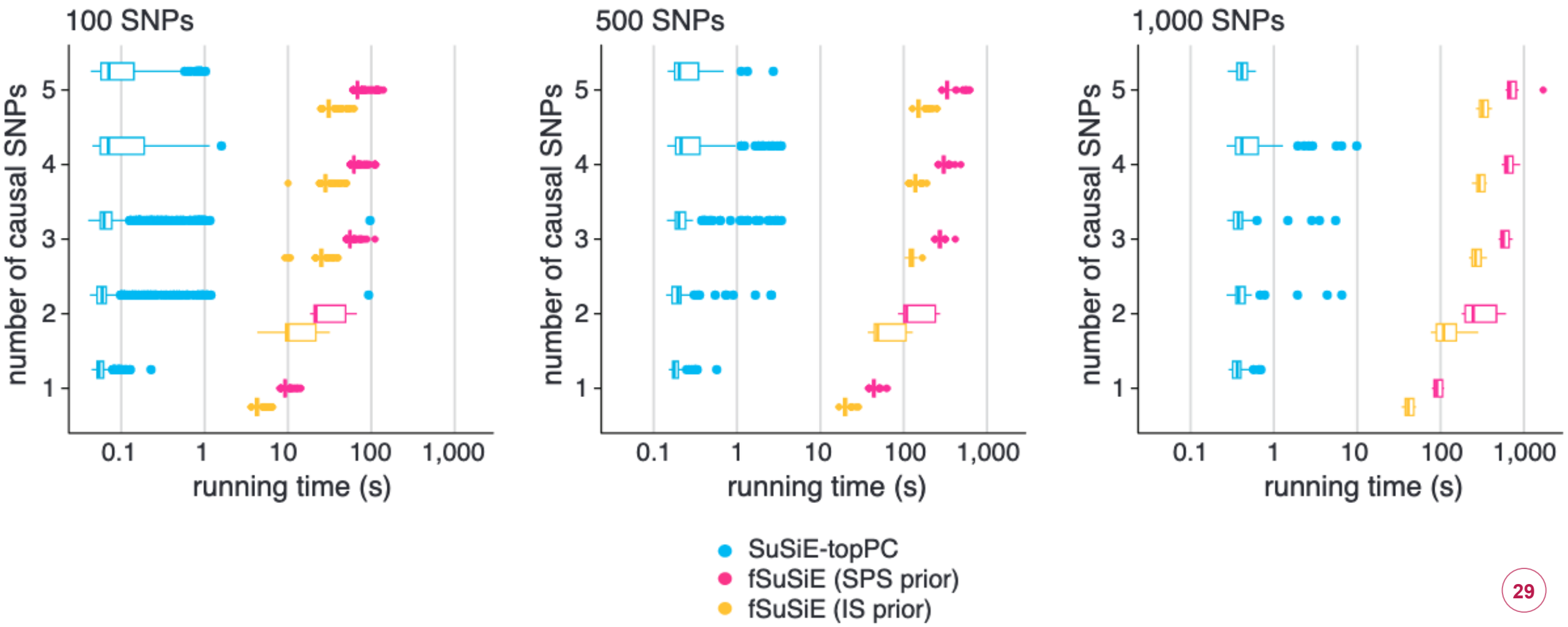
Performance assessment of CS size



Performance assessment of recovery of affected CpGs



Performance assessment of running times

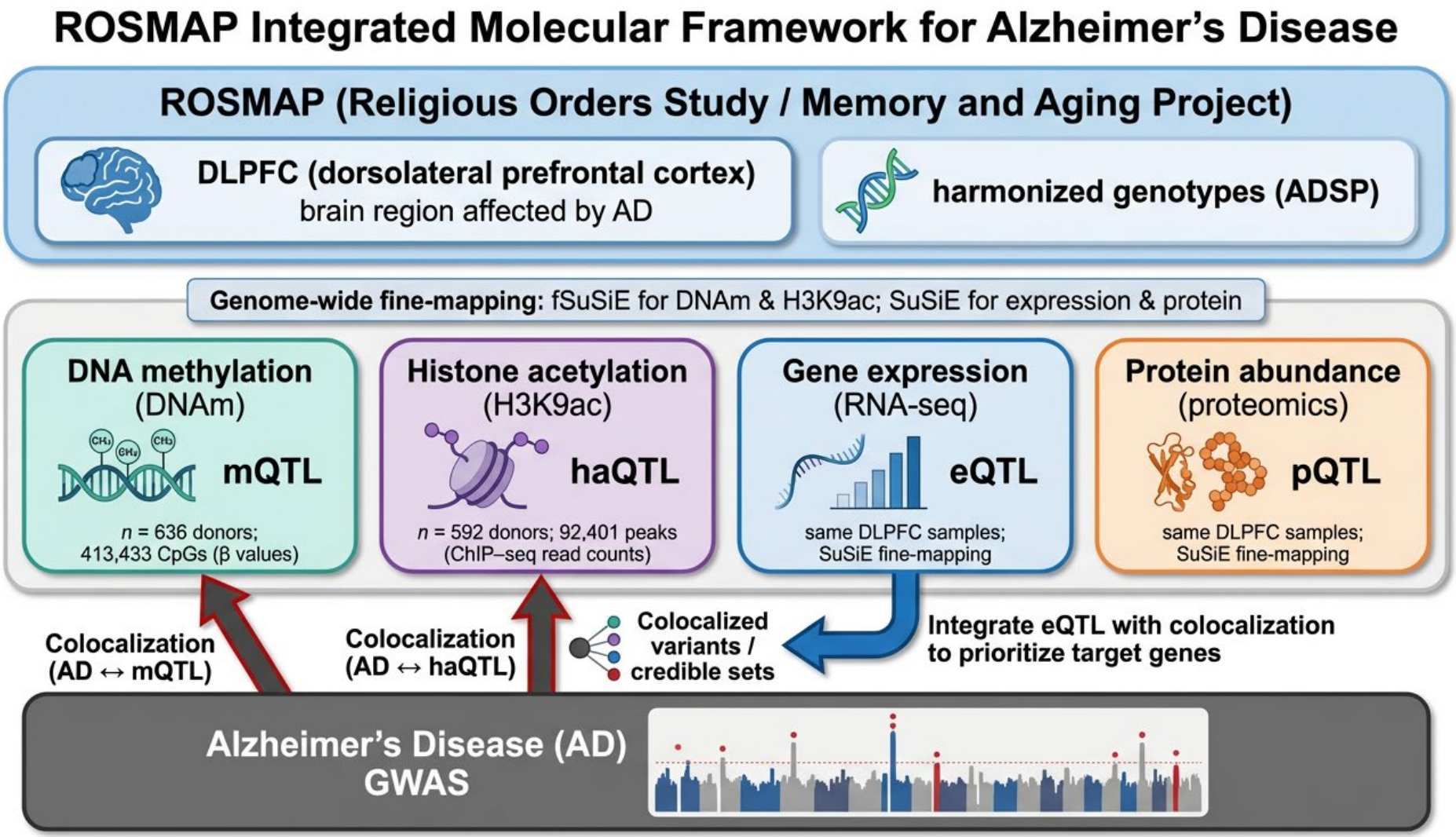


香港城市大學
City University of Hong Kong

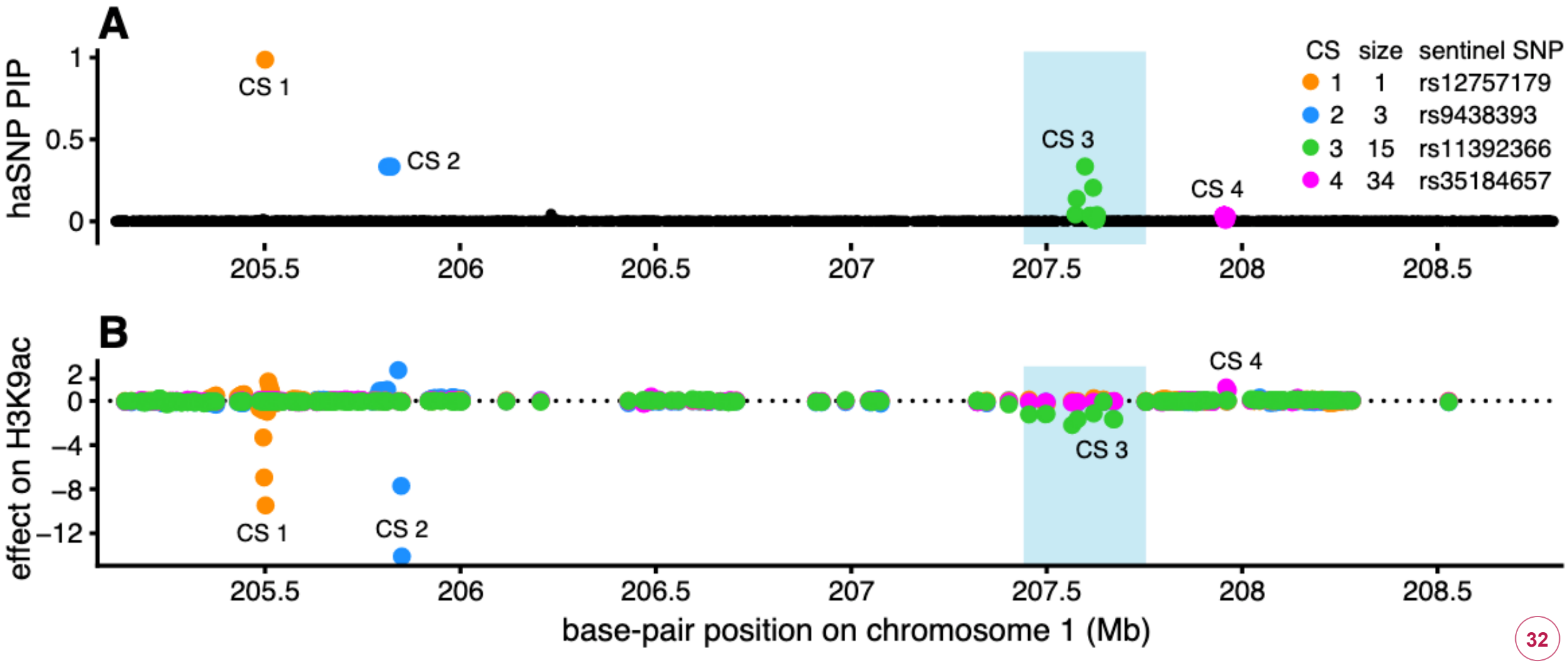
04

Real-data applications

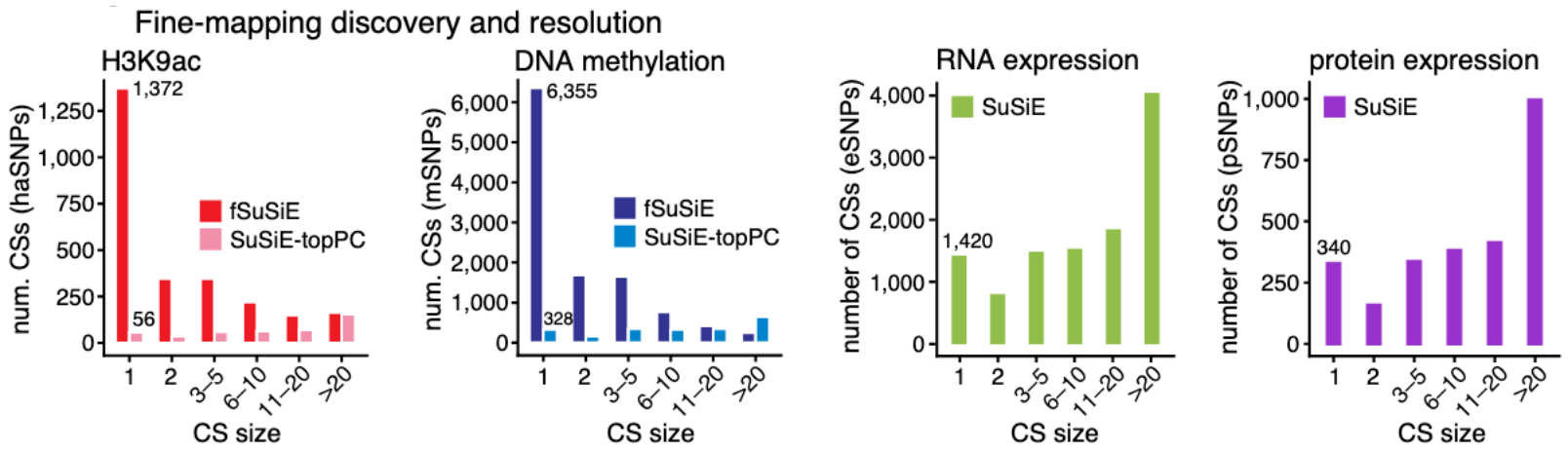
Introduction of Genome-wide fine-mapping analysis

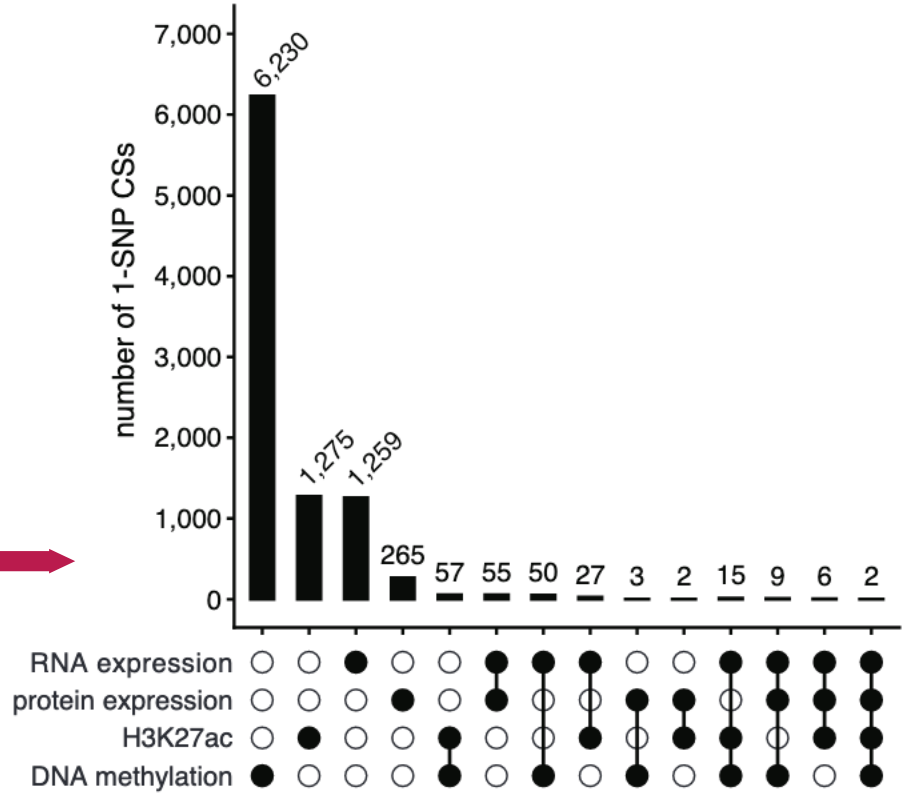


An example of H3K9ac fine-mapping with fSuSiE



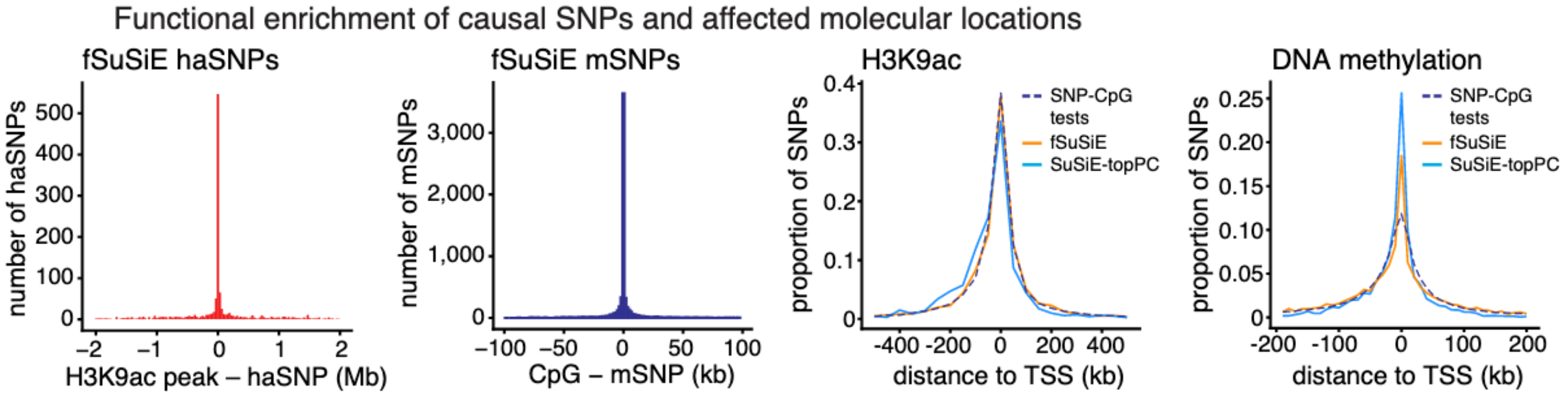
Fine-mapping discovery and resolution



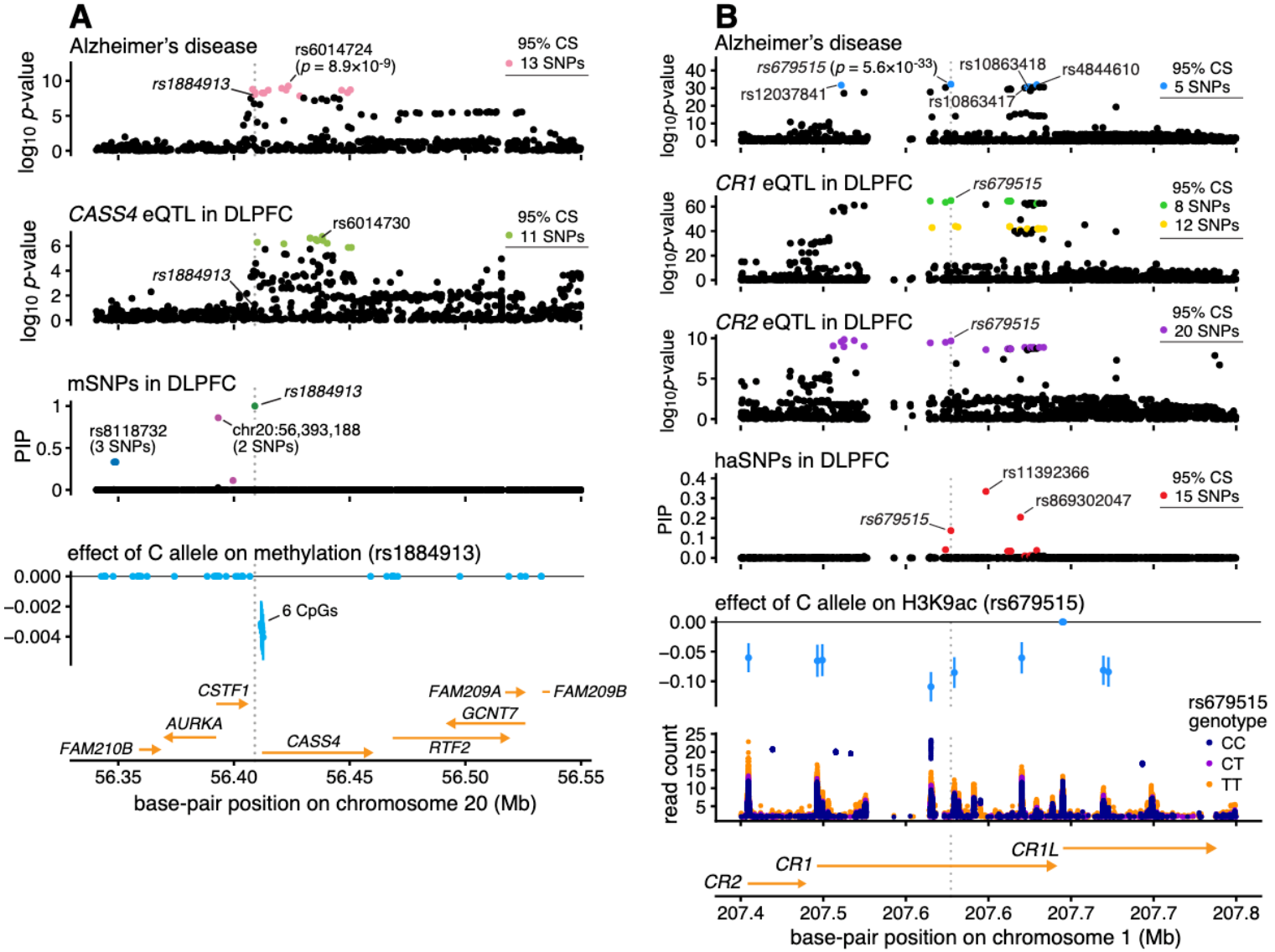


For reference, the total number of single-variant CSs for each trait is:
H3K9ac (1,372), methylation (6,355),
RNA expression (1,420), protein expression (340).

Functional enrichment of causal SNPs and affected molecular location



Colocalization analysis results for AD GWAS summary statistics



香港城市大學
City University of Hong Kong

THANK YOU

Speaker: Yuekai Li

Major: Biostatistics